A Note on a Gap in Case 2 of the previous Lower-Bound Argument for Problem #650

Abstract

This note records a gap in Case 2 of the lower-bound argument given in `650_lowerbound_v1.txt`. The problem is that, after deleting the midpoint $b_0 = x + N$ from the right-hand side, the proof still argues as if the next multiple after the largest multiple in B_- must lie in B_+. This is false. We isolate the incorrect assertion, give an explicit counterexample, and explain which subsequent deductions therefore remain unproved.

1 The Case 2 setup

Let

$$A = \{a_1 < \cdots < a_m\}, \qquad N := a_m, \qquad I = (x, x + 2N),$$

and suppose that

$$x \in N\mathbb{Z}.$$

Then the unique multiple of N in I is

$$b_0 := x + N.$$

Case 2 of the proof introduces

$$A_0 := A \setminus \{N\}, \qquad B_- := (x, x + N) \cap \mathbb{Z}, \qquad B_+ := (x + N, x + 2N) \cap \mathbb{Z},$$

and

$$B_0 := B_- \sqcup B_+ = (I \cap \mathbb{Z}) \setminus \{b_0\}.$$

It then forms the bipartite graph G_0 with left vertex set A_0 and right vertex set B_0, where $a \in A_0$ is joined to $b \in B_0$ if and only if $a \mid b$.

For $S \subseteq A_0$, let $\Gamma_0(S)$ denote the neighbourhood of S in G_0.

2 The incorrect assertion

The proof claims that for every nonempty $S \subseteq A_0$ one may "repeat verbatim" the Case 1 argument as follows: for each $a \in S$, if c_a denotes the largest multiple of a lying in B_-, then

$$c_a + a \in B_+,$$

and hence

$$a \longmapsto (c_a, c_a + a)$$

defines an injection from S into

$$(\Gamma_0(S) \cap B_-) \times (\Gamma_0(S) \cap B_+).$$

This assertion is false.

Proposition 1. *There are valid Case 2 data for which the statement*

$$c_a + a \in B_+$$

fails.

Proof. Take

$$m = 4, \qquad A = \{1, 2, 3, 6\}, \qquad x = 0.$$

Then

$$N = 6, \qquad b_0 = x + N = 6,$$

so we are in Case 2. We have

$$A_0 = \{1, 2, 3\},$$

and

$$B_- = (0, 6) \cap \mathbb{Z} = \{1, 2, 3, 4, 5\}, \qquad B_+ = (6, 12) \cap \mathbb{Z} = \{7, 8, 9, 10, 11\}.$$

Now choose

$$S = \{2\} \subseteq A_0.$$

The largest multiple of 2 in B_- is

$$c_2 = 4.$$

But then

$$c_2 + 2 = 6 = b_0 \notin B_+.$$

Thus the asserted conclusion $c_2 + 2 \in B_+$ is false. □

3 Why this breaks the proof

The failure above is not cosmetic. It invalidates the key map used in Case 2.

Indeed, in the example from the proposition, the neighbourhood of $S = \{2\}$ in G_0 is

$$\Gamma_0(S) = \{2, 4, 8, 10\},$$

so

$$\Gamma_0(S) \cap B_- = \{2, 4\}, \qquad \Gamma_0(S) \cap B_+ = \{8, 10\}.$$

The proof attempts to define

$$\phi(a) := (c_a, c_a + a).$$

For $a = 2$ this gives

$$\phi(2) = (4, 6),$$

but $6 \notin B_0$, hence

$$(4, 6) \notin (\Gamma_0(S) \cap B_-) \times (\Gamma_0(S) \cap B_+).$$

Therefore the proposed map is not even well-defined in general.

Consequently, the argument does *not* establish the inequality

$$|S| \leq |\Gamma_0(S) \cap B_-| \cdot |\Gamma_0(S) \cap B_+|,$$

and so it does *not* establish

$$|\Gamma_0(S)| \geq \lceil 2\sqrt{|S|} \rceil \qquad (\varnothing \neq S \subseteq A_0)$$

in this version of the proof.

But this neighbourhood bound is exactly what is then used in the two remaining parts of Case 2:

- the Hall argument for $m = 4$;
- the dummy-vertex Hall argument for $m \geq 5$.

Hence those subsequent deductions are left without justification.

4 Conceptual explanation of the obstruction

The obstruction is structural. In Case 1, the argument uses the fact that if c_a is the largest multiple of a on the left side of the interval decomposition, then the next multiple $c_a + a$ still lies on the right side of the graph.

In Case 2, however, the midpoint

$$b_0 = x + N$$

is removed from the right-hand vertex set. Therefore it may happen that the next multiple after c_a is exactly the deleted point b_0, not a vertex of G_0.

More precisely, whenever

$$a \mid b_0 \qquad \text{and} \qquad c_a = b_0 - a$$

is the largest multiple of a in B_-, one has

$$c_a + a = b_0 \notin B_+.$$

Thus one cannot simply reuse the Case 1 injection in Case 2.

5 A minor endpoint issue

There is also a minor endpoint inaccuracy in the sentence asserting

$$x + 2N - a < v_a < x + 2N.$$

The left inequality should be written as

$$x + 2N - a \leq v_a < x + 2N,$$

since the largest multiple of a below $x + 2N$ may equal $x + 2N - a$. This point is not the main difficulty; the essential gap is the failure of $c_a + a \in B_+$ discussed above.

6 Conclusion

The conclusion of this note is therefore the following:

> The Case 2 argument in `650_lowerbound_v1.txt` is incomplete. The specific step asserting that the map
>
> $$a \longmapsto (c_a, c_a + a)$$
>
> lands in
>
> $$(\Gamma_0(S) \cap B_-) \times (\Gamma_0(S) \cap B_+)$$
>
> is false, and hence the displayed proof does not establish the required neighbourhood bound in Case 2.

This does *not* show that the lower bound theorem itself is false. It shows only that the Case 2 proof in this particular version requires a genuinely different construction.